RIGHT EXACTNESS OF THE FILTERED-COLIMIT EXTENSION, WITH THE FP-REDUCTION CHAIN EXECUTED THREE TIMES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file functors.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Span recomputation by label pairing (LANE_BRIEF 10.2): in this chapter the author's proof environment follows the statement environment as a standalone block; the spec covers the complete environment pair (statement span, proof span) as recorded in the pool candidate.

Editorial note. Occupation scan: combined claim_key + statement_anchor + generic-line scan over all registration files (active and withdrawn) found 0 hits for functors.tex (fu-occupied.json, 2026-10-14); consistent with the duty manager's six-check dispatch.

Source reference. Original lemma 0GNJ: sources/stacks-chow/tags.txt, source lines 20794–20794. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 0DLP: sources/stacks-chow/tags.txt, source lines 17054–17054. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 0GNK: sources/stacks-chow/tags.txt, source lines 20795–20795. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 0GMN: sources/stacks-chow/tags.txt, source lines 20763–20763. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 00HA: sources/stacks-chow/tags.txt, source lines 636–636. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 002M: sources/stacks-chow/tags.txt, source lines 103–103. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 0519: sources/stacks-chow/tags.txt, source lines 6527–6527. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let A, $\mathcal{B}$, F be as in Lemma 0GNJ. Assume $\mathcal{B}$ is additive, has cokernels, and F is right exact. Then F' is additive, right exact, and commutes with arbitrary direct sums.*

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Proof. Since F is right exact, F commutes with coproducts of pairs, which are represented by direct sums. Hence F is additive by Homology, Lemma 0DLP. Hence F' is additive and commutes with direct sums by Lemma 0GNK. We urge the reader to prove that F' is right exact themselves instead of reading the proof below.

To show that F' is right exact, it suffices to show that F' commutes with coequalizers, see Categories, Lemma 0GMN. Now, if $a, b : K \to L$ are maps of A-modules, then the coequalizer of a and b is the cokernel of $a - b : K \to L$. Thus let $K \to L \to M \to 0$ be an exact sequence of A-modules. We have to show that in

$$F'(K) \to F'(L) \to F'(M) \to 0$$

the second arrow is a cokernel for the first arrow in $\mathcal{B}$ (if $\mathcal{B}$ were abelian we would say that the displayed sequence is exact). Write $M = \operatorname{colim}_{i \in I} M_i$ as a filtered colimit of finitely presented A-modules, see Algebra, Lemma 00HA. Let $L_i = L \times_M M_i$. We obtain a system of exact sequences $K \to L_i \to M_i \to 0$ over I. Since colimits commute with colimits by Categories, Lemma 002M and since cokernels are a type of coequalizer, it suffices to show that $F'(L_i) \to F(M_i)$ is a cokernel of $F'(K) \to F'(L_i)$ in $\mathcal{B}$ for all $i \in I$. In other words, we may assume M is finitely presented. Write $L = \operatorname{colim}_{i \in I} L_i$ as a filtered colimit of finitely presented A-modules with the property that each L_i surjects onto M. Let $K_i = K \times_L L_i$. We obtain a system of short exact sequences $K_i \to L_i \to M \to 0$ over I. Repeating the argument already given, we reduce to showing $F(L_i) \to F(M_i)$ is a cokernel of $F'(K) \to F(L_i)$ in $\mathcal{B}$ for all $i \in I$. In other words, we may assume both L and M are finitely presented A-modules. In this case the module $\operatorname{Ker}(L \to M)$ is finite (Algebra, Lemma 0519). Thus we can write $K = \operatorname{colim}_{i \in I} K_i$ as a filtered colimit of finitely presented A-modules each surjecting onto $\operatorname{Ker}(L \to M)$. We obtain a system of short exact sequences $K_i \to L \to M \to 0$ over I. Repeating the argument already given, we reduce to showing $F(L) \to F(M)$ is a cokernel of $F(K_i) \to F(L)$ in $\mathcal{B}$ for all $i \in I$. In other words, we may assume K, L, and M are finitely presented A-modules. This final case follows from the assumption that F is right exact. $\square$